Clinical Research

VEKARD score: clinical scoring system of pediatric heart disease for screening

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ABSTRACT
BACKGROUND Heart disease in children is frequently diagnosed late in life. This study aimed to establish and validate a scoring system to screen for heart disease in children, based on clinical examinations to accelerate referral.

METHODS This 2-phase cross-sectional study was conducted in pediatric cardiology outpatients and emergency room Ngoerah Hospital, Bali, Indonesia, from January to May 2024 and July to October 2024. Children aged 0–18 who were suspected of having heart disease and willing to be research participants were included in this study. Patients who had already been diagnosed with heart disease and acute lower respiratory tract infection were excluded from the study. The chi-square, multivariate logistic regression, and ROC analysis were performed to identify and validate variables eligible for the scoring system. Phase 1 for generating the score, continued to phase 2 to validate the developed score.

RESULTS Of 310 patients enrolled, 15 were excluded from the study. Congenital heart disease was the dominant condition in this study. The first model of the scoring system, named VEKARD score, consisted of a significant murmur, exercise intolerance, difficulty in breathing, and SpO_2 <95%. The second model replaced SpO_2 <95% with cyanosis, which can be used if pulse oximetry is not available. Using a sensitivity-first threshold suitable for screening purposes, a cutoff ≥1 for model 1 and cutoff ≥2 for model 2 yielded sensitivities of 85% and 81.8%, respectively. Both models underwent validation in a sample of 241 patients. The results demonstrated good discriminative performance, with AUC values of 0.859 for model 1 and 0.853 for model 2.

CONCLUSIONS The VEKARD score can be used to enhance pediatric heart disease screening and accelerate referral to tertiary healthcare facilities, thereby preventing delays in treatment.

KEYWORDS children, heart disease, heart murmur, prediction, score

Heart disease, congenital heart disease (CHD), and acquired heart disease (AHD) are major problems in children owing to delayed detection and diagnosis. Although newborn screening for critical CHD is available, its implementation remains limited, and many cases of pediatric heart disease continue to be diagnosed late.[1] Delayed diagnosis is often caused by late referrals of heart disease cases from primary healthcare, especially in Indonesia, which increases mortality.[2–4] Previous studies by Izhar et al[5] and Alexander Nada have identified a scoring system to detect CHD, which involved clinical examinations and included radiologic and electrocardiographic assessments as part of the criteria.[5,6] These scoring systems rely on diagnostic tools that are not consistently accessible, especially in primary healthcare facilities in Indonesia.

To our knowledge, a scoring system for heart disease in children has not been developed yet, particularly in Indonesia. Despite the existence of critical CHD screening as a routine government program, its implementation has been suboptimal due to limited facilities and the availability of primary and secondary healthcare services. The VEKARD score is a screening tool to facilitate pediatric heart disease screening, which was developed in two models, designed to accommodate settings where pulse oximetry is either available or unavailable. Therefore, this study aimed to develop and validate a heart disease scoring system for screening children to facilitate referrals.

METHODS

Study design

This cross-sectional study was conducted in two phases. Phase 1 aimed to develop a scoring system to facilitate screening and referral for heart disease in children. Phase 2 continued to validate the scoring system. This study was conducted in a pediatric cardiology outpatient and pediatric emergency room at Ngoerah Hospital, Bali, Indonesia, a tertiary hospital responsible for receiving patients, as a referral center for the East Indonesia region. Tertiary hospitals provide the most complete diagnostic tools, allowing a comprehensive assessment of the patients; therefore, scoring systems need to be developed. Phase 1 was from January to May 2024 and phase 2 was from July to October 2024.

Patient selection

The study population consisted of pediatric inpatients and outpatients aged 0–18 years with suspected heart disease. Children referred for evaluation and whose parents provided consent were included. The exclusion criteria were prior diagnosis or treatment of heart disease and the presence of an acute lower respiratory tract infection. The study conformed to the ethical principles of the Declaration of Helsinki, for medical research involving human subjects.

The minimum required sample size for phase 1 was 155, calculated using the rule-of-thumb formula.[7] Consecutive sampling methods were used for data collection. Data were screened and analyzed using the SPSS software version 29 (IBM Corp., USA). Phase 2 was conducted to validate the developed pediatric heart disease screening score. The sample size calculation based on AUC assumed a 5% alpha error, a 10% beta error, and an expected AUC of 0.80, resulting in a minimum required sample of 99 participants.[7] A second calculation based on sensitivity was performed using an expected sensitivity of 80%, a 5% alpha error, a 10% beta error, and the previously reported prevalence of pediatric heart disease of 46%, yielding a minimum required sample of 135 participants.[7,8]

Parameters assessed

In the phase 1, the assessed variables included: (1) significant murmur, defined as a systolic murmur of grade ≥3/6 or any diastolic murmur, which assessed by pediatric residents and validated by pediatric cardiologist; (2) cyanosis, characterized by bluish discoloration of the tongue, lips, or skin with peripheral oxygen desaturation; (3) failure to thrive, defined as length-for-age or weight-for-age below two standard deviations from the mean for at least 56 days (in infants ≤5 months) or 3 months (in infants >5 months); (4) exercise intolerance, indicated by easy fatigability during activities, presenting as tachypnea or tachycardia in children or poor feeding in infants; (5) tachycardia, defined as a heart rate above the age-specific normal; (6) difficulty in breathing, increased respiratory rate (tachypnea) above normal limit for age with or without increased work of breathing (retractions); (7) peripheral oxygen saturation (SpO2) <95% using pulse oximetry; (8) heart disease, either CHD or AHD, diagnosed using a standardized echocardiography as the gold standard, performed by a pediatric cardiologist with more than 15 years' experience.

Statistical analysis

Scoring was conducted using bivariate analysis (chi-square) to obtain the odds ratios (ORs) of the independent variables. Variables with $p<0.25$ in bivariate analysis were entered into multivariate logistic regression (backward stepwise), and those with $p<0.05$ were retained in the scoring models.[9] Quality of this predictive model was analyzed using the Hosmer–Lemeshow test, and receiver operating characteristic (ROC) analysis was performed to obtain the area under the curve (AUC) of the model. A well-calibrated model was defined if the Hosmer–Lemeshow test was >0.05, and an appreciable screening performance was defined as an AUC ≥80%.

The cutoff point was determined by considering the sensitivity-first threshold. This measure identifies the threshold that represents the point at which the scoring system achieves high sensitivity while preserving an acceptable level of specificity for clinical screening.[10]

The validation process included screening accuracy measures based on the AUC and a 2 × 2 crosstab analysis to estimate the sensitivity, specificity, negative predictive value (NPV), and likelihood ratios (LRs), along with their corresponding confidence intervals (CIs). As this tool is intended for screening, the analysis emphasizes high-sensitivity operating thresholds and screening-oriented metrics. Phase 2 was conducted to validate the pediatric heart disease scoring model. The sample size calculation based on the AUC, with 5% alpha error, 10% beta error, and expected AUC, was 0.80, resulting in a minimum required sample of 99. Observer bias was minimized by the data being blinded to the examiners and the echocardiography results were validated by an experienced pediatric cardiologist. This study was approved by the Research Ethics Committee of the Faculty of Medicine, Universitas Udayana, Bali, Indonesia (No: 2022/UN14.2.2.VII.14/LT/2024).

RESULTS

Of 310 patients, 15 ineligible patients were excluded (9 patients had already been diagnosed and treated for heart disease and 6 patients had lower respiratory tract infections). The characteristics of the 295 patients included in this study are shown in Table 1. Most patients (63%) were referred from primary or secondary healthcare facilities. Echocardiography was performed on all patients to diagnose heart disease. Most acyanotic CHD cases involve patent ductus arteriosus, ventricular septal defects, and atrial septal defects. Tetralogy of Fallot was the most common cyanotic CHD, followed by transposition of the great arteries and double outlet right ventricles. Comorbidities observed with CHD included anal and esophageal atresia, clinical Down syndrome, congenital talipes equinovarus, labiognatopalatoschizis, multiple congenital anomalies, microtia, Dandy–Walker variant, tracheomalacia, gastroschisis, omphalocele, polydactyly, congenital hypothyroidism, thalassemia, systemic lupus erythematosus (SLE), cholestasis, malignancy, and HIV infection, which were found in 16.6% of participants (Table 1).

Bivariate analysis was performed to analyze the association between independent variables and heart disease in children (Table 2). Two models were

Table 1. Patient characteristics of children evaluated in phase 1 and 2

Variables	Phase 1, n (%) (N = 295)	Phase 2, n (%) (N = 241)
Age (years), median (min–max)	1 (0–17)	0 (0–17)
0–1	156 (52.9)	147 (61.0)
2–5	34 (11.5)	22 (9.2)
Male gender	150 (50.8)	95 (39.4)
Referral source		
Primary healthcare	13 (4.4)	7 (2.9)
Secondary healthcare	157 (53.3)	203 (84.2)
In Ngoerah hospital*	10 (3.4)	31 (12.9)
Non-Bali	6 (2.0)	0 (0)
Nutritional status†		
Well-nourished	210 (71.2)	185 (76.8)
Protein energy malnutrition		
Mild–moderate	61 (20.7)	34 (14.1)
Severe	24 (8.1)	22 (9.1)
Stunted†	112 (38.0)	76 (31.5)
Chest pain	12 (4.1)	20 (8.3)
Syncope	4 (1.4)	0 (0)
Fever	24 (8.1)	12 (5.0)
Family history of heart disease (1st degree)‡	3 (1.0)	3 (1.2)
Comorbid	49 (16.6)	29 (12.0)
Arrhythmia	16 (5.4)	11 (4.6)
Hypertension	3 (1.0)	1 (0.4)
Hepatomegaly	16 (5.4)	4 (1.7)
Edema	16 (5.4)	9 (3.7)
Type of heart disease		
CHD		
Acyanotic	153 (51.9)	127 (52.7)
Cyanotic	34 (11.5)	20 (8.3)
AHD		
RHD	26 (8.8)	9 (3.7)
Kawasaki disease	2 (0.7)	1 (0.4)
Other acquired§	5 (1.7)	1 (0.4)
Not heart disease	75 (25.4)	83 (34.4)
Complications		
Heart failure	14 (4.7)	11 (4.6)
PAH	22 (7.5)	4 (1.7)
IE	6 (2.0)	1 (0.4)
Combined	11 (3.7)	7 (2.9)

AHD=acquired heart disease; CHD=congenital heart disease; IE=infective endocarditis; PAH=pulmonary arterial hypertension; RHD=rheumatic heart disease

*Indicate cases referred to pediatric cardiology (outpatient/inpatient) via the internal division at Ngoerah hospital; †nutritional status and stunted were defined using World Health Organization (WHO) criteria, as follows: stunted=height-for-age z-score <−2 (WHO), protein-energy malnutrition severity per WHO/other source; ‡first-degree family history=parents or siblings; §other acquired=myocarditis, myxoma developed based on the availability of pulse oximetry in the healthcare facilities. The first model included SpO2 <95% as a variable and the second model replaced SpO2 <95% with cyanosis. The results of multivariate analysis of the models were statistically significant, and the scoring calculation for each variable is shown in Table 3. The probability of heart disease can be predicted based on the total score using the formula (Equation 1):

$$\left[\frac{1}{1+\exp(-y)}\right] \qquad (1)$$

Table 2. Associations between clinical signs and echocardiography-confirmed heart disease in children (phase 1)

Variables	Heart disease Yes, n (%)	Heart disease No, n (%)	*p*	OR (95% CI)
Failure to thrive*	74 (33.6)	15 (20.0)	**0.026**	2.027 (1.079–3.811)
Significant murmur	127 (57.7)	6 (80.0)	**<0.001**	15.704 (6.540–37.712)
Cyanosis	37 (16.8)	2 (2.7)	**0.002**	7.380 (1.734–31.414)
Exercise intolerance	74 (33.6)	4 (5.3)	**<0.001**	8.997 (3.163–25.589)
Tachycardia	42 (19.1)	8 (10.7)	0.093	1.976 (0.882–4.427)
Difficulty in breathing	108 (49.1)	9 (12.0)	**<0.001**	7.071 (3.357–14.894)
SpO2 <95%	56 (25.5)	3 (4.0)	**<0.001**	8.195 (2.483–27.049)

CI=confidence interval; OR=odds ratio; SpO2=peripheral oxygen saturation
*World Health Organization (WHO) weight-for-age z-score <–2

with the regression equation obtained from total score logistic regression analysis: [y = −0.431 + 1.384 total score] for the first model and [y = −0.353 + 0.815 total score] for the second model. The AUC of the first and second models were 0.857 (*p*<0.001) (95% CI: 0.810–0.904) (Figure 1a) and 0.848 (*p*<0.001), respectively (95% CI: 0.799–0.897) (Figure 1b).

The scoring system developed was named the VEKARD score. The score for each variable and probability of heart disease were presented as scoring cards that could be used in primary and secondary healthcare facilities, as shown in Supplementary Tables 1 and 2. The first model of the VEKARD score demonstrated an appreciable determinant value with a cutoff point of 1, yielding a sensitivity of 85% (95% CI: 79.58–89.44), specificity of 77.3% (95% CI: 66.21–86.21), positive predictive value (PPV) of 91.6% (95% CI: 87.83–94.37), NPV of 63.74% (95% CI: 55.63–71.13), and accuracy of 83.05% (95% CI: 78.27–87.15). Similarly, the second model of the VEKARD score also showed an acceptable determinant value with a cutoff point of 2, which had a sensitivity of 81.8% (95% CI: 76.08–86.68), specificity of 81.3% (95% CI: 70.67–89.40), PPV of 92.78% (95% CI: 88.87–95.39), NPV of 60.40% (95% CI: 53.03–67.32), and accuracy of 81.69% (95% CI: 76.80–85.94) (Table 4).

Validation of the VEKARD score models was conducted during phase 2, with a total of 241 eligible patients. The patients' characteristics are described in Table 1. Most referred to Bali (80.9%), followed by West Nusa Tenggara, East Nusa Tenggara, Java, and Sulawesi. Clinical Down syndrome was

Table 3. Multivariate logistic regression and point-score calculation of the 1st and 2nd models

Model	Sig.	Exp(B)	Lower	Upper	B	S.E.	Score
The first model							
Significant murmur	**<0.001**	12.007	4.825	29.881	2.49	0.47	2
Exercise intolerance	**0.002**	6.011	1.952	18.506	1.79	0.57	1
Difficulty in breathing	**0.001**	3.960	1.733	9.050	1.38	0.42	1
SpO2 <95%	**0.006**	6.070	1.687	21.832	1.80	0.65	1
The second model							
Significant murmur	**<0.001**	10.967	4.427	27.167	2.39	0.46	3
Cyanosis	0.091	3.902	0.803	18.957	1.36	0.81	1
Exercise intolerance	**0.001**	6.222	2.042	18.955	1.83	0.57	2
Difficulty in breathing	**<0.001**	4.381	1.944	9.874	1.48	0.42	2

B=coefficient; Exp(B)=odds ratio; SE=standard error; SpO2=peripheral oxygen saturation
X=B divided by SE (B/SE); Y=X divided by the lowest value of X among all variables. Scoring was determined by rounding the Y value, where values ≥0.5 were rounded up and <0.5 were rounded down

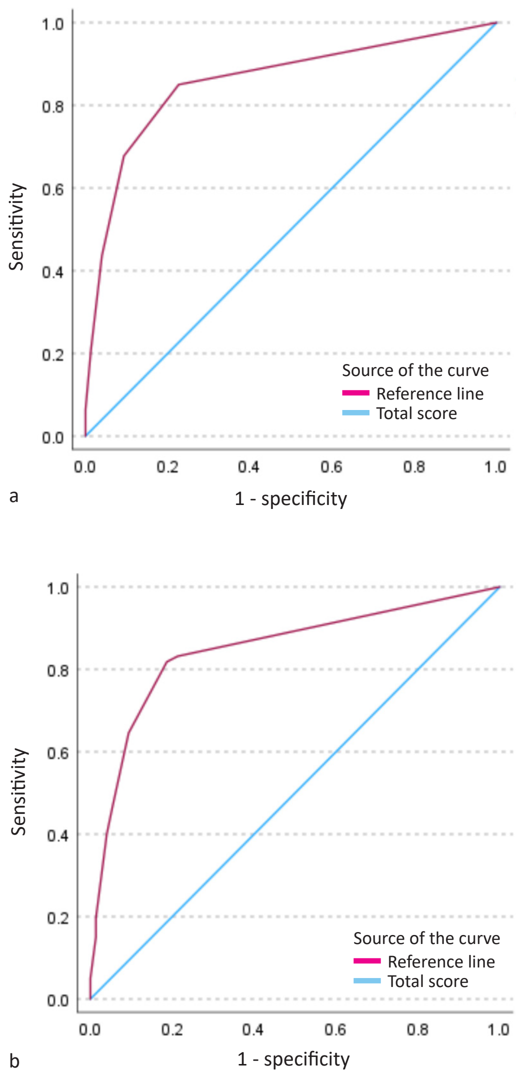

Figure 1. Receiver operating characteristic (ROC) curves for predicting heart disease in children. (a) First model; (b) second model

predominantly found as a comorbid condition (41.4%), and other comorbidities were epilepsy, renal abnormalities, anorectal malformation, congenital hypothyroidism, HIV, labiognatopalatoschizis, congenital rubella syndrome, SLE, thanatophoric dysplasia, and cryptorchidism. Of these patients, 9.1% had complications such as heart failure, infective endocarditis, pulmonary arterial hypertension, or both.

CHD was dominant in this study (61.0%), comprising 52.7% acyanotic and 8.3% cyanotic CHD. AHD was found in 4.6% of patients, and 34.4% of patients had no heart disease. All patients underwent echocardiography as a diagnostic tool for heart disease, and 158 of 241 patients (65.6%) were diagnosed with heart disease. The screening performance of VEKARD score models 1 and 2 are described in Table 5.

The first model of the scoring system, which used pulse oximetry, had favorable screening performance with a sensitivity of 89.87% (95% CI: 84.08–94.10), specificity of 81.93% (95% CI: 71.95–89.52), positive LR of 4.97 (95% CI: 3.14–7.89), negative LR of 0.12 (95% CI: 0.08–0.20), PPV of 90.45% (95% CI: 85.65–93.75), NPV of 80.95% (95% CI: 72.54–87.24), and accuracy of 87.14% (95% CI: 82.24–91.09).

The second model of the scoring system, without pulse oximetry, also had a similar screening performance, with a sensitivity of 88.61% (95% CI: 82.59–93.11), specificity of 81.93% (95% CI: 71.95–89.52), positive LR of 4.90 (95% CI: 3.09–7.78), negative LR of 0.14 (95% CI: 0.09–0.22), PPV of 90.32% (95% CI: 85.47–93.67), NPV of 79.07% (95% CI: 70.74–85.52), and accuracy of 86.31% (95% CI: 81.31–90.38).

According to the ROC analysis, both models had values similar to those of phase 1. The first model had an AUC value of 0.859 (95% CI: 0.804–0.914, $p<0.001$), which was similar to the second model, which had an AUC value of 0.853 (95% CI: 0.797–0.909, $p<0.001$). Both scoring systems were statistically categorized as robust models based on the AUC (80–90%), showing that these scoring systems can be implemented for pediatric heart disease, considering the availability of pulse oximetry in healthcare facilities.

DISCUSSION

The VEKARD score is a screening tool to facilitate pediatric heart disease screening, which in this study was developed in two models. The first model of the VEKARD score consisted of significant murmur, exercise intolerance, difficulty in breathing, and SpO_2 <95% variables, whereas the second model replaced the SpO_2 <95% variable with cyanosis. Therefore, the second model can be used as an alternative if pulse oximetry is not available in healthcare facilities.

The murmur can be examined during physical examination with heart auscultation. Murmurs occur because of changes in blood flow or turbulence resulting from mechanical alterations in the valve, septum, or heart outflow. Murmur was found to be prominent in heart valve abnormalities, and 61.1% of patients underwent echocardiography because of the presence of murmur.[11–14] This study found that pathological murmur was related to heart disease,

Table 4. Cutoff sensitivity and specificity of model 1 and 2

Cutoff	Model 1		Model 2	
	Sensitivity	Specificity	Sensitivity	Specificity
0.5000	**85.0**	**77.3**	83.2	78.7
1.5000	67.7	90.7	**81.8**	**81.3**
2.5000	43.6	96.0	64.5	90.7
3.5000	20.9	98.7	46.4	94.7
4.5000	6.4	100.0	40.0	96.0
6.0000	0.0	100.0	19.5	98.7
6.5000	-	-	15.0	98.7
7.5000	-	-	5.0	100.0
9.0000	-	-	0.0	100.0

Table 5. Chi-square analysis of model 1 and 2 of the VEKARD score

Variables	Heart disease*		*p*	OR (95% CI)
	Yes, n (%)	No, n (%)		
Model 1				
Score ≥1	142 (90.4)	15 (9.6)	**<0.001**	40.23 (18.79–86.15)
Score <1	16 (19.0)	68 (81.0)		
Model 2				
Score ≥2	140 (90.3)	15 (9.7)	**<0.001**	35.26 (16.76–74.19)
Score <2	18 (20.9)	68 (79.1)		

CI=confidence interval; OR=odds ratio
*Echocardiography-confirmed congenital/acquired structural heart disease

similar to a previous CHD scoring system that was validated in Indonesia.[5,6,15,16]

This study found that exercise intolerance was dominant in heart disease cases, similar to a study that reported decreased exercise tolerance, particularly in cyanotic CHD.[17] The exercise intolerance occurred as a heart failure symptom in children and adolescents. Exercise intolerance may present as poor feeding in infants due to excess pulmonary circulation, leading to diaphoresis and cyanosis during feeding.[18] In children, exercise intolerance may manifest as lower exercise capacity than in healthy children.[19,20]

Difficulty in breathing included increasing respiratory rate, chest wall retractions, and nasal flaring found in heart disease patients, especially children with heart failure.[21] Heart disease affects respiratory function chronically, including fluid retention in the lung, lung edema, increased lung pressure due to cardiomegaly, hemodynamic changes, increased risk of respiratory tract infection, pulmonary hypertension, thrombosis, or bleeding.[22] Difficulty in breathing was found to be dominant in patients with heart disease in this study.

SpO2 is a simple yet essential method used for critical CHD screening in newborns[21] with SpO2 <95% indicating low blood oxygen levels (oxygen desaturation), which could be a sign of heart disease in children. Cyanosis, a bluish discoloration of the tongue, lips, conjunctiva, or nails, usually occurs if SpO2 is <85%, and is subjective based on the examiner's experience and clinical judgement. Cyanosis can also be caused by lung or neurological abnormalities, which require further examination.[23] Therefore, pulse oximetry is recommended as a simple objective method to indicate low oxygen levels in the blood.

Cyanosis is commonly found in cyanotic CHD, acyanotic CHD, or AHD with severe pulmonary hypertension, lung abnormalities, and cerebral abnormalities and is rarely found in blood or liver abnormalities. The prevalence of cyanotic CHD is estimated to be 25% of all cases of CHD. Cyanosis is caused by connection abnormalities between the heart chamber and/or heart circulation system, and is divided into early- and late-onset cyanosis based on its pathology. Patients with cyanotic CHD are often found to have tachypnea, difficulty in breathing, lung edema, tachycardia, and murmur.[21] Cyanosis has been reported to be dominant in CHD patients. A previous study reported that cyanosis with an SpO2 cutoff of 80% had a sensitivity of 79–95% and a specificity of 72–95% for the detection of severe hypoxemia. This examination detected cyanotic CHD in neonates (92–96%), similar to that in our study.[24]

Some VEKARD score variables were similar to those in Nada's criteria. The Nada's criteria were validated in 60 patients with CHD in North Sumatra, Indonesia. It used a cutoff of 3 with an accuracy of 87.14%, a sensitivity of 88.3%, a specificity of 80%, and an AUC value of 0.883.[6] Nada's criteria included a combination of clinical manifestations and some supporting examinations, such as chest X-ray and electrocardiography abnormalities, as its criteria for diagnosing CHD. In contrast, the VEKARD score is based only on clinical manifestations, without supporting examinations, to enhance its feasibility, especially in limited healthcare facilities. These VEKARD score models can be used to screen for pediatric heart disease, whether congenital or acquired, with a screening performance similar to that of Nada's criteria for CHD.

In Japan, a heart disease screening system was established in 1954, and electrocardiography has been used since 1995. Heart disease screening was conducted in schools to screen students for cardiovascular abnormalities using a screening form, electrocardiography, and phonocardiography. However, its cost-effectiveness still needs to be considered.[25] In Indonesia, electrocardiography, pulse oximetry, and expertise are not equally available in all healthcare facilities. Therefore, these scoring systems provide a simple method for detecting heart disease in children based on physical examinations with or without pulse oximetry.

Another study conducted in India used clinical scoring to detect CHD in neonates. The scoring system uses historical and physical examinations. Historical screening consisted of bluish discoloration, difficulty in feeding and breathing, increased precordial activities or pulsations, failure to thrive, cyanotic spells, a history of consanguinity, and a family history of CHD. The physical examination findings used in the scoring system included the clinical stigmata of specific syndromes, pulse oximetry abnormalities, blood pressure or pulse abnormalities, precordium abnormalities, hepatomegaly, abnormal heart sounds, and murmurs. The screening test was performed on 970 patients, most of whom had failure to thrive, difficulty in feeding, and difficulty in breathing. The most common signs were murmur, followed by tachycardia and desaturation. The cutoff point was 3, with sensitivity of 96.77%, specificity of 98.72%, PPV of 71.43%, NPV of 99.89%, and accuracy of 98.66%.[5] Two models of VEKARD score that established by this study had been validated in the phase 2 of this study, resulted similar and suitable screening performances.

The VEKARD score models are clinical scoring systems that can be implemented to enhance pediatric heart disease screening without using any supporting examinations, such as chest radiography and electrocardiography, and can accelerate referral to prevent delayed treatment. These scoring systems included two models that could be used according to pulse oximetry availability, both of which have been validated with good screening performance. This scoring system is intended for screening, especially in primary and secondary healthcare facilities (which are often limited in echo or trained professionals), to identify probable heart disease cases before referral to advanced healthcare facilities. However, this scoring system cannot replace echocardiography as the gold standard for diagnosing heart disease. Because this scoring system is intended for screening, it uses a high sensitivity threshold to minimize false-negative results. Although this approach may lead to more referrals and greater use of resources, the main priority in resource-limited settings is to avoid missing children who may have heart disease. Therefore, there is no comparison of sensitivity/specificity between scoring *versus* echocardiography, as scoring is for screening and echocardiography is for diagnosis.

The limitations of the study include the fact that the researchers did not stratify CHD and AHD, and the validation of the VEKARD score was conducted in a referral center (tertiary hospital). This study may have resulted in observer bias, as echocardiography examinations for diagnosing heart disease were performed by pediatric residents. To minimize bias, all examinations were reassessed and validated by an experienced pediatric cardiologist. External validation was not performed using a larger population. Therefore, further investigations are needed, especially in primary and secondary healthcare facilities.

The VEKARD scoring system consists of two models that facilitate heart disease screening in children. The first model consisted of significant murmur, exercise intolerance, breathing difficulty, and SpO_2 <95%. The second model replaces SpO_2 <95% with cyanosis, which can be used without pulse oximetry, particularly in rural areas. Both models were validated with good determinant values and can be used to enhance the referral process for children with heart disease.

Conflict of Interest

The authors affirm no conflict of interest in this study.

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